COAST: Coordinated Ownership and Anticipatory Sequential Routing for Dynamic Vehicle Routing with Time Windows

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Abstract

Dynamic Vehicle Routing Problems with Time Windows (DVRPTW) require sequential routing policies to make decisions that are simultaneously feasible in the short term, coordinated across multiple vehicles, and effective over long planning horizons under dynamic customer arrivals. Existing neural routing policies typically compress these heterogeneous objectives into a single compatibility score, making coordination and anticipation difficult to learn jointly. We propose COAST (Coordinated Ownership and Anticipatory Sequential Routing), a neural routing framework that reformulates customer selection as a decision decomposition problem. Specifically, COAST estimates three complementary decision signals: Edge-aware Compatibility for immediate routing feasibility, Ownership Memory for implicit fleet coordination, and Candidate-conditioned Lookahead for anticipatory decision making. A learned nonlinear score-fusion MLP combines these signals into the routing policy, optimized end-to-end through reinforcement learning without auxiliary supervision. Extensive experiments demonstrate consistent improvements over classical and neural baselines, together with strong cross-scale transfer and stable performance across controlled dynamic operating conditions. Ablation studies confirm that the three decision signals provide complementary contributions. These results suggest that explicitly decomposing routing decisions into complementary reasoning components provides an effective framework for dynamic vehicle routing.

1 Introduction

The Vehicle Routing Problem (VRP) is a fundamental combinatorial optimization problem with broad applications in logistics, transportation, and last-mile delivery. While classical VRP assumes that all customer requests are known in advance (Toth and Vigo 2014), many real-world systems operate in dynamic environments where requests arrive continuously and routing decisions must be updated online. This setting is modeled as the Dynamic Vehicle Routing Problem (DVRP) (Psaraftis 1988).

Among DVRP variants, the Dynamic Vehicle Routing Problem with Time Windows (DVRPTW) is particularly challenging because routing decisions must satisfy time-window and capacity constraints while future customer arrivals remain unknown. Traditional approaches, including dispatching rules, rolling-horizon optimization, and periodic re-optimization, often require repeated optimization and substantial problem-specific engineering. Deep reinforcement learning (DRL) has emerged as an attractive alternative by learning routing policies directly from data, enabling fast online decision making without repeatedly solving optimization problems.

Despite recent progress, existing neural routing policies still struggle to coordinate multiple vehicles while anticipating the long-term consequences of online decisions. Most methods rely on a shared policy to simultaneously capture fleet-level coordination and long-horizon planning, forcing these distinct reasoning processes into a single decision mechanism. Consequently, policy robustness may deteriorate when customer arrivals, spatial distributions, or time-window characteristics differ from those encountered during training.

To address this limitation, we propose **COAST** (**Co**ordinated **O**wnership and **A**nticipatory **S**equential Rou**T**ing), a reinforcement learning framework that explicitly separates coordination and long-term planning into complementary decision signals. COAST integrates an edge-aware compatibility module for local vehicle–customer interactions, a memory-based ownership mechanism for fleet coordination, and a candidate-conditioned lookahead signal for long-horizon planning. A learned nonlinear score-fusion MLP combines these signals into a unified routing policy trained end-to-end with reinforcement learning.

The main contributions of this paper are summarized as follows:

- We present a structured decision decomposition framework for DVRPTW that explicitly separates fleet coordination from long-term planning within sequential routing policies.
- We introduce a memory-based ownership mechanism that enables implicit vehicle coordination without explicit inter-vehicle communication.
- We introduce a candidate-conditioned anticipatory scoring head that provides action-specific evidence optimized through long-horizon policy returns.

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- Extensive experiments demonstrate consistent improvements over classical and neural baselines, together with strong cross-scale transfer and stable performance across controlled dynamic operating conditions.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 formulates the problem. Section 4 presents the proposed COAST framework. Section 5 reports experimental results, and Section 6 concludes the paper.

2 Related Work

Deep reinforcement learning (DRL) has emerged as a competitive paradigm for neural combinatorial optimization in VRPs (Nazari et al. 2018). Early work used recurrent encoder–decoder policies to learn routing heuristics directly from data (Nazari et al. 2018), while attention-based Transformers substantially improved representation learning and solution quality across VRP variants (Kool, Van Hoof, and Welling 2018). More recent studies enhance scalability, generalization, and expressive power through transferable local policies (Gao et al. 2023), heterogeneous attention (Wang et al. 2025), lightweight cross-attention (Luo et al. 2025), edge-aware Transformers (Meng et al. 2025), and diverse decoding strategies (Hottung, Mahajan, and Tierney 2025). MVMoE and RouteFinder further target multi-task generalization across static VRP variants (Zhou et al. 2024; Berto et al. 2025a), while RL4CO supports standardized and reproducible evaluation of neural combinatorial optimization methods (Berto et al. 2025b). Despite these advances in representation, search, diversity, and transfer, routing decisions are typically generated by a single vehicle–customer compatibility function that entangles local feasibility, fleet coordination, and long-term routing considerations.

For DVRP, reinforcement learning is an attractive alternative to repeatedly solving an optimization model online. Existing studies formulate the problem as a Markov decision process and learn dispatching policies with value-based or actor–critic reinforcement learning (Akkerman, Mes, and van Jaarsveld 2025; Konovalenko and Hvattum 2024). Recent work incorporates evolving request information into dynamic policies (Konovalenko and Hvattum 2024; Bono et al. 2021), while graph and edge-aware encoders improve structured routing representations (Meng et al. 2025). Although these advances improve policy quality and generalization, decisions remain primarily driven by a unified compatibility score, making it difficult to distinguish immediate feasibility from the long-term consequence of selecting each customer.

Another fundamental challenge is coordinating multiple vehicles in dynamic environments. Multi-agent routing policies address this through event-driven sequential decisions, shared attention, or learned coordination mechanisms (Bono et al. 2021; Zhang et al. 2020; Shahbazian et al. 2025). While such strategies improve cooperation, they generally require repeatedly modelling the fleet during decision making, increasing computational cost and limiting scalability. Furthermore, standard actor–critic methods estimate a state-value function for variance reduction. Because it is independent of the selected action, the critic provides identical guidance to all candidate customers in a state and cannot explicitly assess their different downstream routing consequences.

Taken together, existing learning-based routing policies exhibit three important limitations: (i) local feasibility, fleet coordination, and long-term planning are entangled within a single compatibility score; (ii) effective vehicle coordination often relies on computationally expensive fleet-level interaction; and (iii) state-value critics provide action-independent guidance that cannot explicitly assess the long-term impact of individual routing decisions. To address these limitations, we propose COAST, which decomposes customer selection into three complementary signals: an edge-aware compatibility module for local feasibility, a memory-based ownership mechanism for implicit fleet coordination, and a candidate-conditioned lookahead signal for action-specific anticipatory evidence. A learned nonlinear score-fusion MLP combines these signals into a unified routing policy trained end-to-end using reinforcement learning.

3 Problem Statement

Dynamic VRPTW. A problem instance contains depot 0 and customer requests $\mathcal{C} = \{1, \dots, L_c\}$. Each request j has normalized two-dimensional position $\mathbf{p}_j = (p_{j,x}, p_{j,y})^\top \in [0,1]^2$, demand q_j, time window $[e_j, l_j]$, service duration σ_j, and reveal time ρ_j:

$$\mathbf{x}_j = (\mathbf{p}_j, q_j, e_j, l_j, \sigma_j, \rho_j).$$

At decision event t, the active vehicle k_t is available at physical time $\eta_t = t^{\mathrm{cur}}_{k_t}$. Thus the policy observes only the revealed and unserved requests

$$\mathcal{C}_t = \{j \in \mathcal{C} : \rho_j \le \eta_t\}, \qquad \mathcal{U}_t = \mathcal{C}_t \setminus \mathcal{S}_t,$$

where $\mathcal{S}_t$ is the served set. Vehicle k has state $\mathbf{s}^t_k = (\mathbf{p}^t_k, q^{\mathrm{rem}}_k, t^{\mathrm{cur}}_k)$, and s_t denotes the policy-visible event state comprising the fleet, revealed-request status, and active vehicle. For candidate j, let

$$\begin{aligned} d_{k_t j} &= \|\mathbf{p}^t_{k_t} - \mathbf{p}_j\|_2, \\ b_{k_t j} &= \max\left(t^{\mathrm{cur}}_{k_t} + d_{k_t j}/v, e_j\right), \\ \ell_{k_t j} &= [b_{k_t j} - l_j]_+. \end{aligned}$$

Here $b_{k_t j}$ is the service-start time, including any wait before e_j. We use soft time windows: late service is penalized, rather than masked, as in multi-vehicle soft-window routing (Zhang et al. 2020; Bono et al. 2021).

Event-driven sequential process. Following the event-driven sequential multi-agent Markov decision process (sMMDP) construction of Bono et al. (2021), only one homogeneous vehicle acts at each event. With $\mathcal{D}_t$ denoting completed vehicles,

$$\begin{aligned} k_t &= \arg\min_{k \notin \mathcal{D}_t} t^{\mathrm{cur}}_k, \\ \mathcal{A}_t &\equiv \mathcal{A}(s_t, k_t) = \{0\} \cup \{j \in \mathcal{U}_t : q_j \le q^{\mathrm{rem}}_{k_t}\}. \end{aligned}$$

The depot is always eligible. The mask excludes unrevealed, served, capacity-ineligible, and completed-vehicle actions; it does not exclude late customers. After a_t, the environment updates travel, waiting, service, capacity, customer status,

newly visible requests, and the next earliest-available vehicle. If the depot is selected while requests remain hidden, the simulator advances to the next reveal event instead of terminating.

Objective. The shared policy $\pi_\theta(a_t \mid s_t, k_t)$ minimizes a shaped cost comprising travel distance, tardiness, and pending requests:

$$J(\pi_\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_t (d_{k_t,a_t} + \lambda_{\text{late}}\ell_{k_t,a_t} + \zeta_t\lambda_{\text{pend}}N_t)\right].$$

Here $\zeta_t = 1$ only when the transition triggers the simulator's fleet-completion check (i.e., all vehicles are marked inactive), and is zero otherwise; N_t is the number of unserved requests at that check. Selecting the depot makes the active vehicle inactive. If the completion check occurs while unrevealed requests remain, the simulator advances to the next reveal event and reactivates a vehicle rather than terminating the episode.

4 Methodology

COAST addresses this event-level gap. Unlike task-level expert selection or decoding multiple full solutions, it decomposes evidence among candidates inside each online decision. As illustrated in Figure 1, the event-driven state and eligible-action set are encoded into three complementary signals: local edge-aware compatibility, fleet-relative ownership, and candidate-conditioned lookahead.

4.1 Dynamic representations and local compatibility

At a dynamic decision event, a customer can be globally attractive yet unsuitable for the current vehicle because of its specific travel, timing, and remaining-capacity trade-off. COAST therefore separates context shared among customers from the active vehicle–candidate relation, allowing local selection to respond to the current routing state rather than relying on node similarity alone.

The GraphEncoder produces distance-aware contextual customer embeddings $H_t^{\text{cust}} = \{\mathbf{c}_{t,j}\}$; in the policy forward path, it is recomputed only when newly revealed requests change $\mathcal{C}_t$. At every event, the ActiveVehicleEncoder cross-attends from $\mathbf{s}_{k_t}^t$ to these embeddings to obtain $\mathbf{h}_t^{\text{act}}$. For each active-vehicle–candidate pair, EdgeFeatureEncoder maps the eight dynamic features—distance, travel time, arrival, waiting, lateness, slack, eligibility, and capacity gap—to $\mathbf{e}_{t,k_t,j}^{\text{emb}}$, collected in H_t^{edge}.

CrossEdgeFusion forms the local compatibility score by adding an edge-dependent bias to each attention head:

$$S_{t,j}^{\text{att}} = \frac{1}{H}\sum_{h=1}^{H}\left[\frac{(\mathbf{h}_t^{\text{act}}W_Q^h)(\mathbf{c}_{t,j}W_K^h)^\top}{\sqrt{d_h}} + (\mathbf{w}_b^h)^\top\mathbf{e}_{t,k_t,j}^{\text{emb}}\right].$$

This score captures the immediate vehicle–customer fit without applying a candidate softmax prematurely.

4.2 Ownership and anticipatory signals

Local compatibility alone does not prevent several vehicles from favoring overlapping sets of customers. Ownership gives each candidate a fleet-relative preference by comparing the acting vehicle's evolving memory with the memories of its peers; it provides lightweight persistent coordination rather than repeated all-vehicle message passing at every decision.

COAST maintains a memory tensor M_t with one slot per vehicle. After selecting j^*, only the acting slot changes:

$$M_{t+1}[k_t] = \tanh\big(W_{\text{in}}[\mathbf{h}_t^{\text{act}}, \mathbf{c}_{t,j^*}, \mathbf{e}_{t,k_t,j^*}^{\text{emb}}] + W_{\text{hid}}M_t[k_t]\big),$$
$$M_{t+1}[i] = M_t[i] \quad (i \neq k_t).$$

The OwnershipHead computes bilinear vehicle–customer logits and turns them into the active vehicle's relative claim:

$$o_{t,i,j} = \frac{(W_{\text{veh}}M_t[i])^\top(W_{\text{cust}}\mathbf{c}_{t,j})}{\sqrt{D}},$$
$$S_{t,j}^{\text{owner}} = \log\text{softmax}_{i\in\mathcal{I}}(o_{t,i,j})|_{i=k_t}.$$

Thus ownership is a learned coordination bias across vehicles, not explicit vehicle-to-vehicle message passing. Current fit and relative ownership can still leave candidates with very different downstream routing options. The lookahead head therefore supplies a separate action-conditioned signal from the current vehicle, candidate, and edge state; it learns useful distinctions through long-horizon return feedback.

The LookaheadHead produces an action-conditioned latent routing score,

$$S_{t,j}^{\text{look}} = \text{MLP}_{\text{look}}([\mathbf{h}_t^{\text{act}}, \mathbf{c}_{t,j}, \mathbf{e}_{t,k_t,j}^{\text{emb}}]),$$

which is optimized through long-horizon policy returns rather than acting as a calibrated value or Q-function.

4.3 Adaptive fusion and learning

The three signals capture different aspects of a decision and can have different scales as the eligible set changes. A fixed combination may overemphasize one signal in a regime where another is more informative, so COAST standardizes them within the current eligible set before applying learned nonlinear score fusion. The fusion MLP learns candidate-dependent nonlinear interactions among the normalized signals, rather than assigning explicit state-conditioned weights.

Let $\mathcal{R} = \{\text{att}, \text{look}, \text{owner}\}$ denote the three signals in the left-to-right order of Figure 1. COAST normalizes each $S_{t,j}^r$, for $r \in \mathcal{R}$, only over $j \in \mathcal{A}_t$, then fuses the resulting three-vector with an MLP:

$$\widetilde{S}_{t,j}^r = \frac{S_{t,j}^r - \mu_t^r}{\sqrt{(\sigma_t^r)^2 + \varepsilon}}, \qquad r \in \mathcal{R},$$
$$\mathbf{z}_{t,j} = [\widetilde{S}_{t,j}^{\text{att}}, \widetilde{S}_{t,j}^{\text{look}}, \widetilde{S}_{t,j}^{\text{owner}}],$$
$$S_{t,j}^{\text{final}} = \phi_{\text{fuse}}(\mathbf{z}_{t,j}).$$

The illustrated vectors are $\mathbf{S}_t^{\text{att}} = (S_{t,j}^{\text{att}})_{j\in\mathcal{A}_t}$, $\mathbf{S}_t^{\text{look}} = (S_{t,j}^{\text{look}})_{j\in\mathcal{A}_t}$, and $\mathbf{S}_t^{\text{owner}} = (S_{t,j}^{\text{owner}})_{j\in\mathcal{A}_t}$. The masked policy is

$$\pi_\theta(j \mid s_t, k_t) = \frac{\exp(S_{t,j}^{\text{final}})}{\sum_{u\in\mathcal{A}_t}\exp(S_{t,u}^{\text{final}})}, \qquad j \in \mathcal{A}_t.$$

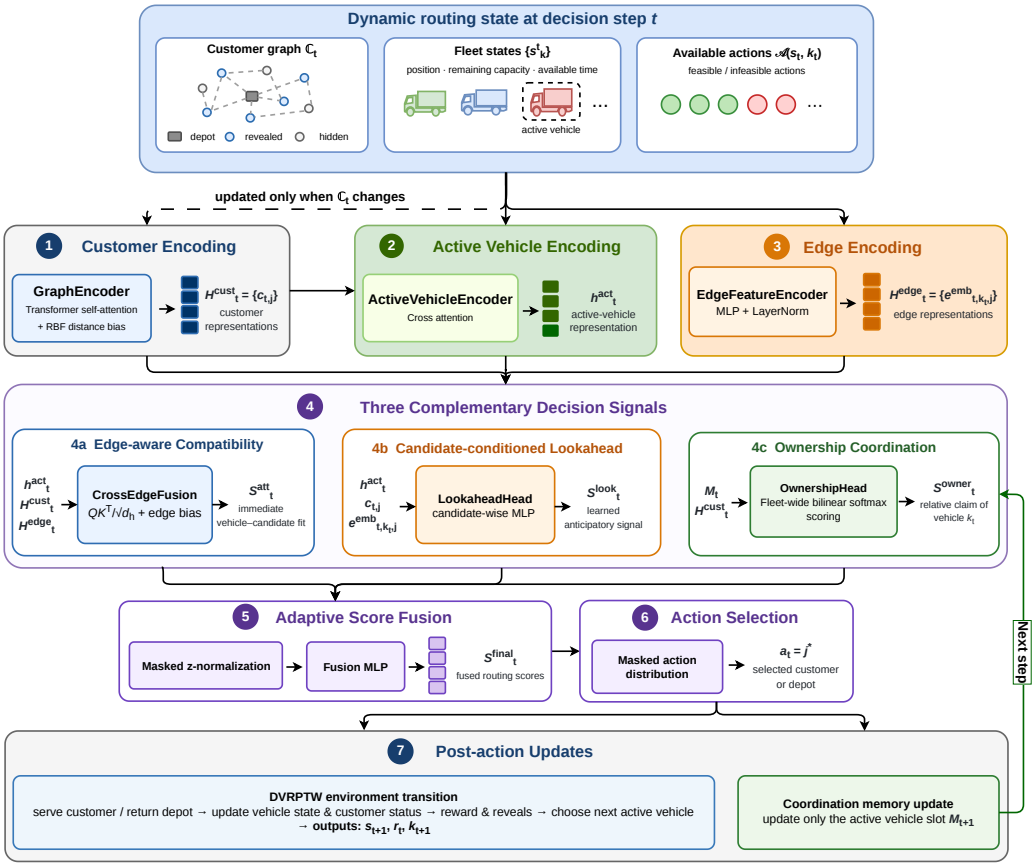


Figure 1: Overview of COAST at decision event t. The policy encodes the observed routing state and nonlinearly fuses edge-aware compatibility, candidate-conditioned lookahead, and ownership signals to select the next action.

Training samples from this distribution and optimizes REINFORCE (Williams 1992) with a learned critic baseline, advantage normalization, and entropy regularization; reported inference uses greedy $\arg\max$ decoding. The ownership and lookahead modules have no auxiliary targets and receive learning signal only through the policy objective.

4.4 Complexity Analysis

Let L_c, L_v, and D denote customer count, vehicle count, and embedding dimension. The Transformer customer encoder dominates with $O(L_c^2 D)$, as in existing attention-based routing models. The Active Vehicle Encoder, compatibility, lookahead, and fusion add $O(L_c D)$; ownership adds $O(L_v L_c D)$. Since $L_v \ll L_c$, the total complexity remains $O(L_c^2 D)$.

5 Experimental results

5.1 Evaluation protocol

We report the normalized objective from Section 3 combining distance, lateness, and unserved-request penalties (lower is better). All neural methods use greedy decoding on fixed test instances. The Solomon-style suite (Solomon 1987) contains 176 instances across `h100`, `h200`, and `h400`; PYTH contains 10,240 instances at each scale. COAST and the internal variants use the same instance generator and routing objective, with each ablation altering the components specified in the RQ3 analysis below. MARDAM (Bono et al. 2021) is the direct event-driven comparator; AM (Kool, van Hoof, and Welling 2019) is the canonical attention-based policy, and PolyNet (Hottung, Mahajan, and Tierney 2025) is a recent learned-diversity method. The GPHH comparison suite (Tam et al. 2026) additionally provides GPHH and the C+C, C+W, and WIQ+C dispatching baselines.

Training configuration. COAST is trained on `n50m3` instances (50 customers, 3 vehicles; horizon 480) with time-window prevalence, dynamic-arrival rate, and early-reveal ratio each sampled from $\{0.25, 0.50, 0.75, 1.00\}$, $\{0.10, 0.25, 0.50, 0.75\}$, and $\{0, 0.50, 0.75, 1.00\}$, respectively. The policy uses 128-dimensional embeddings, two 4-head attention layers, feed-forward width 256, 8-dimensional edge features, and 128-dimensional memory and lookahead states (0.764M trainable parameters). REINFORCE with a learned critic uses Adam with policy/critic learning rates $10^{-4}/10^{-3}$, entropy coefficient 0.01, gradient clipping at 2.0, and batches of 512, for 500 epochs of 1,000 updates over a reshuffled 512,000-instance pool. Training used an

Table 1: Solomon-style normalized routing cost across all eight compared methods (lower is better). "Poly." denotes PolyNet. COAST is the best neural method in 8 of the 9 size–topology aggregates; MARDAM is lower only on h100-C. COAST beats GPHH on 172/176 instances and each dispatching rule on all 176

Scale	Top.	COAST	MARDAM	AM	Poly.	GPHH	C+C	C+W	WIQ+C
h100	C	67.7	**65.1**	73.0	70.4	80.6	130.7	118.1	175.0
	R	**57.2**	60.0	66.1	64.0	72.0	147.7	143.1	179.9
	RC	**65.7**	67.4	73.1	70.3	80.1	153.0	141.1	181.7
h200	C	**117.6**	162.4	126.6	137.0	191.7	269.6	248.7	361.2
	R	**104.2**	142.7	124.0	125.7	184.9	276.6	273.0	349.0
	RC	**105.2**	143.5	125.7	130.5	181.0	282.9	283.1	351.0
h400	C	**220.7**	323.8	248.4	258.8	374.9	630.1	616.1	752.3
	R	**202.0**	314.4	244.0	238.6	354.6	624.1	618.7	720.1
	RC	**203.3**	322.7	250.3	245.0	352.7	634.8	636.8	724.2

NVIDIA T4 GPU (16 GB). Results use greedy decoding from one validation-selected checkpoint minimizing mean cost on 10,240 instances.

Statistical protocol. On the same 176 Solomon-style instances, a Friedman rank test (Friedman 1937) compares all eight methods, followed by two-sided paired Wilcoxon signed-rank tests (Wilcoxon 1945) with Holm correction (Holm 1979) for the seven planned COAST–baseline contrasts. A neural-only Friedman test provides a like-for-like complement.

5.2 RQ1: Does COAST improve benchmark routing quality?

Question. Does the decision decomposition improve routing quality across spatial topologies, dispatching baselines, and increasing fleet scale?

Table 1 reports all eight methods across the R, C, and RC Solomon topologies.

Solomon benchmark. The Solomon table establishes three points. First, COAST is best in every h200 and h400 size–topology aggregate. Second, the small-scale exception is explicit rather than averaged away: at h100, MARDAM is 3.8% lower on the compact C topology, while COAST still leads the h100 aggregate and its R and RC subsets. Measured from the scale-level means, COAST's reduction over the best neural comparator is 1.9%, 13.0%, and 15.1% at h100, h200, and h400, respectively.

Third, the same table places the neural comparison in a broader dispatching context. COAST beats GPHH on 172/176 individual instances and every dispatching rule on all 176. The separation is especially pronounced at h400: on R instances, COAST's 201.98 cost is below GPHH's 354.62 and the two savings-based rules' 624.08 and 618.70 (Clarke and Wright 1964). Thus, the result is not limited to one learned comparator, while the neural baselines remain the more demanding comparison because they are much closer.

Table 2(a) shows where the Solomon gains arise. C is consistently the closest topology: it has the lowest COAST win rate and median reduction for each of the three neural comparisons. Even there, COAST wins 82.5% of matched instances against MARDAM and 87.7% against PolyNet. The R and RC results are stronger, with 89.3–98.4% win rates across the three neural comparators. The spatial evidence therefore does not depend on a single topology.

PYTH benchmark. Table 2(b) examines joint customer–fleet scaling on a fixed generated family (10,240 matched instances/row). Ownership has no effect with one vehicle, explaining the smallest gap. With multiple vehicles, COAST improves on the best neural competitor by 5.5%, 4.7%, 7.2%, and 6.1% from n50m3 through n400m20. Although non-monotone in scale, the gap persists after every joint customer–fleet increase, consistent with a coordination benefit rather than a simple route-length effect.

Statistical validation. Table 3 reports the Solomon statistical validation. Friedman tests reject equal performance for all eight methods, the neural subset, and each R, C, and RC topology; Holm-adjusted Wilcoxon tests confirm that COAST significantly outperforms every reported baseline across matched instances.

5.3 RQ2: How do dynamism and time windows change the advantage?

Question. Does COAST retain an advantage as customer arrivals become more dynamic and time windows become tighter?

We evaluate a full-factorial n50m3 grid with four DoD levels, four TW ratios, and 32 matched instances per cell (512 total). This is a *within-support sensitivity analysis*: the factors occur in data construction, so it is not an out-of-distribution result. Figure 2 reports all four neural methods with one shared colour scale, so cellwise comparison is not restricted to one baseline.

Across all 16 cells in Figure 2, COAST has the lowest cost and MARDAM is next-lowest, making it the most stringent cellwise comparator; AM and PolyNet remain higher. At DoD $= 0.10$, COAST's reduction over MARDAM is 0.5–1.5% across TW ratios; at DoD $= 0.75$, it is 2.3–6.2%. The largest reduction occurs at DoD $= 0.75$, TW $= 0.25$, and remains 5.6% and 5.1% at TW $= 0.50$ and 0.75, respectively. Across the full grid, COAST's reduction ranges from 5.4–13.0% versus AM and 4.4–13.9% versus PolyNet. The high-dynamism pattern is therefore not an artifact of selecting one baseline; it is consistent with the greater cost of early decisions when more requests remain to arrive.

The interaction is not monotonic in both factors. At the widest TW ratio (1.00), COAST still leads every comparator, but its high-DoD reduction over MARDAM falls to 2.3%. Thus, the evidence does not support a claim that lookahead or ownership helps uniformly in every condition. Instead, within this grid, the largest COAST margins occur under high dynamism when time windows are not widest. This pattern is consistent with, but does not itself prove, the proposed separation of current vehicle–customer compatibility from fleet coordination and future impact.

The marginal means reinforce the cellwise result. All methods become costlier as TW increases, yet COAST remains nearly flat from DoD $= 0.10$ to 0.50 (43.64 to 43.61) before rising to 44.45 at 0.75; MARDAM increases monotonically from 44.07 to 46.68. At TW $= 1.00$, COAST reaches

Table 2: Detailed RQ1 benchmark evidence. In (a), "win / $\tilde{\Delta}$" gives COAST's instance win rate and median paired percentage cost reduction against each neural comparator. In (b), each PYTH scale has 10,240 matched instances and reports mean normalized cost under greedy decoding

Topology	n	MARDAM win / $\tilde{\Delta}$	AM win / $\tilde{\Delta}$	PolyNet win / $\tilde{\Delta}$
R	63	95.2% / 23.1%	98.4% / 15.4%	93.7% / 14.8%
C	57	82.5% / 21.3%	89.5% / 9.6%	87.7% / 11.0%
RC	56	89.3% / 25.0%	94.6% / 15.8%	92.9% / 15.0%

(a) Topology-stratified neural comparisons.

PYTH scale	COAST	MARDAM	AM	PolyNet
n20m1	**24.78**	26.13	25.40	25.58
n50m3	**43.92**	46.48	49.58	49.93
n100m5	**94.85**	99.50	103.85	105.13
n200m10	**182.38**	203.24	196.53	200.89
n400m20	**355.99**	430.48	378.98	389.72

(b) PYTH greedy-decoding results.

Table 3: RQ1 statistical validation on the 176 matched Solomon instances. (a) Friedman tests. (b) Holm-adjusted Wilcoxon post hoc tests comparing COAST with each baseline.

Scope	n	χ^2_F	p_F
All 8 methods	176	1081.14	3.52e−229
Neural subset	176	244.03	1.28e−52
R (all 8)	63	397.49	8.23e−82
C (all 8)	57	336.30	1.05e−68
RC (all 8)	56	352.19	4.19e−72

(a) Friedman tests.

Baseline	W/L	$\tilde{\Delta}$	$T_{\min}$	p_{raw}	p_{Holm}
MARDAM	157/19	23.4%	522	7.03e−27	7.03e−27
AM	166/10	14.4%	206	4.03e−29	1.21e−28
PolyNet	161/15	13.0%	408	1.12e−27	2.24e−27
GPHH	172/4	27.6%	27	1.97e−30	8.68e−30
C+C	176/0	62.7%	0	1.24e−30	8.68e−30
C+W	176/0	62.5%	0	1.24e−30	8.68e−30
WIQ+C	176/0	68.1%	0	1.24e−30	8.68e−30

(b) Two-sided Wilcoxon tests.

Table 4: Fusion and structural controls: percentage increase in mean cost over COAST. B5 preserves all signals but fixes their fusion to an equal-weight sum. n20m1 is omitted because it has no fleet coordination

Control	n50m3	n100m5	n200m10	n400m20
B5: fixed fusion	+3.6%	+7.2%	**+42.1%**	**+44.9%**
B0: base attention	+3.1%	+2.6%	+2.3%	+2.1%
B1: memory-only	+3.4%	+3.0%	+2.1%	+1.5%
B3: lookahead-only	+2.3%	+2.1%	+1.3%	+0.5%

45.90, compared with 46.62 for MARDAM and approximately 50.7 for AM and PolyNet. Relative to MARDAM, COAST therefore has a flatter response to arrival dynamism while retaining an advantage throughout the time-window sweep.

5.4 RQ3: Which components matter, and when?

Question. Which decision signals account for COAST's gains, and do their effects become clearer in the regimes they were designed to address?

COAST has lower mean cost in all 28 variant–scale comparisons. Figure 3 contains only the three direct removals: each line is the mean-cost increase after removing one named signal. Removing ownership increases cost by 1.7–2.9% at every scale, removing edge features by 2.5–3.4%, and removing lookahead by 0.3–1.9%. The direct-removal view shows a consistent scale effect for ownership and edge features; the smaller lookahead effect motivates the targeted regime analysis below rather than a universal claim. B5 is excluded from this panel because it changes the fusion rule rather than removing a signal.

Table 4 addresses a separate question: whether the signals require learned nonlinear fusion. B5 retains the three signal sources but replaces the fusion MLP with an equal-weight sum. Its cost increase grows from 3.6% at n50m3 to 7.2%, 42.1%, and 44.9% at the three larger scales. Because B5 changes fusion rather than removing a source of information, it tests whether nonlinear score interactions are beneficial. B0, B1, and B3 show that simpler score compositions are also consistently worse; because each changes several connections at once, they are structural controls rather than source-specific causal tests.

Figure 4 asks *when* the designs matter. In its first three rows, a positive cell means that removing the named signal increases cost; in the B5 row, it means that replacing learned nonlinear fusion with fixed fusion increases cost. Removing ownership increases cost in every targeted regime (2.3–3.5%), consistent with its coordination role. Lookahead has its largest effects under burst arrivals (+5.3%) and tight windows (+5.2%). The targeted analysis is consistent with an anticipatory role, particularly under burst arrivals and restrictive temporal regimes. Edge features have their largest effects under tight windows (+4.0%) and sparse spatial structure (+4.7%), where travel and temporal-feasibility cues are more discriminative.

Negative and near-neutral cells are reported rather than masked by an average. Removing lookahead improves the sparse-spatial regime by 2.2%, and fixed-fusion B5 is near-neutral or favorable in some targeted regimes despite its large degradation on the scale benchmark. These patterns concern distinct evaluation axes: Table 4 varies the standard PYTH scale, whereas Figure 4 perturbs operating characteristics.

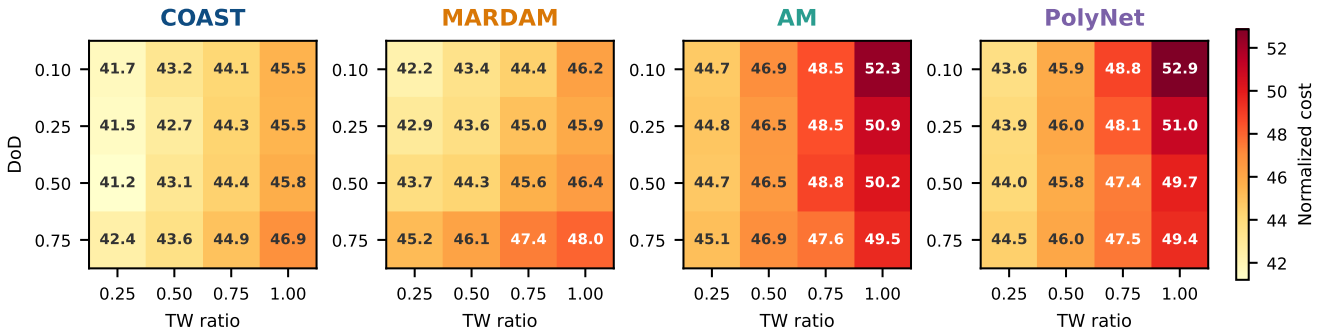


Figure 2: Cellwise DoD–TW comparison of all four neural methods arranged horizontally. Each panel displays the mean normalized cost of one method over 32 matched instances per cell; lower is better. All panels share a common colour scale, enabling direct visual comparison across methods. COAST is lowest in every cell, while MARDAM is the closest comparator throughout this grid.

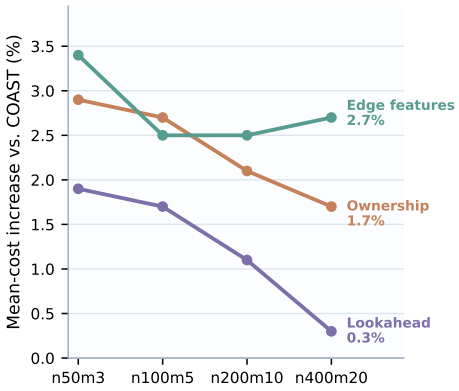


Figure 3: Direct signal removals across scale. Each line reports the mean-cost increase after removing one named COAST signal.

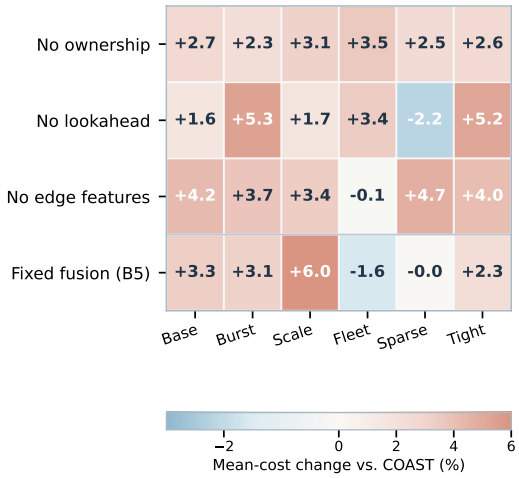


Figure 4: Targeted operating-regime effects. Each cell reports the percentage mean-cost change over 500 matched instances; positive (warm) values favor retaining the original design.

Together, they support a more precise conclusion: ownership, anticipation, and edge-level feasibility are complementary context-sensitive signals, while learned nonlinear fusion is important for the multi-vehicle scale transfer tested here. They do not establish that each signal helps in every possible operating condition.

Limitations: The lookahead head is optimized only through the policy objective and is not calibrated as a value estimator. Moreover, the current evaluation focuses on normalized aggregate routing cost, while a finer behavioral analysis of distance, lateness, and service completeness would provide additional insight. Finally, the study considers homogeneous fleets and soft time windows; extending COAST to heterogeneous vehicles and hard temporal constraints remains future work.

6 Conclusions and Future works

This paper presented **COAST**, a reinforcement learning framework for DVRPTW that reformulates customer selection as a decision decomposition through three complementary signals: Edge-aware Compatibility, Ownership Memory, and Candidate-conditioned Lookahead. Experiments demonstrated consistent improvements over classical and neural baselines, strong cross-scale transfer, and robust performance across controlled dynamic conditions. Ablation results supported the overall effectiveness of the decomposition, although the incremental benefit of lookahead varied across operating regimes.

Future work will investigate lightweight auxiliary supervision and calibration-aware fusion to improve the reliability of lookahead. We also plan to conduct finer-grained behavioral analyses of distance, lateness, and service completeness, and extend COAST to heterogeneous fleets, hard time windows, pickup-and-delivery settings, and stochastic travel times.

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